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Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative
Marketing

Median-of-Means Two-Stage Least Squares Regression

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Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

1 Preliminaries

1.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \quad (1)$$

where $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term. We impose the following conditions.

(A1) **Exogeneity.** $\mathbb{E}[Z_i \varepsilon_i] = 0$.

(A2) **Relevance.** $\mu_{ZX} := \mathbb{E}[Z_i X_i] \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZY}^2 := \text{Var}(Z_i Y_i) < \infty$, $\sigma_{ZX}^2 := \text{Var}(Z_i X_i) < \infty$, and $\sigma_{Z\varepsilon}^2 := \text{Var}(Z_i \varepsilon_i) < \infty$.

Define $\mu_{ZY} := \mathbb{E}[Z_i Y_i]$. By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY} / \mu_{ZX}$.

Remark 1.1 (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products $Z_i Y_i$, $Z_i X_i$, and $Z_i \varepsilon_i$. Thus, no assumptions are imposed on the marginal distributions of Y_i , X_i , Z_i , or ε_i .*

Remark 1.2 (Relationship between variances). *The three variances in (A3) are related. Since $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}. \quad (2)$$

To see this, set $U = Z_i Y_i$ and $V = \beta Z_i X_i$, so that $U - V = Z_i \varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}$.

1.2 Sub-Gaussian Estimators

Definition 1.3 (Sub-Gaussian Estimators). *Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{\mu}(X)$ denote an estimator for μ . For some constant $L > 0$, $\bar{\mu}(X)$ is said to be a sub-Gaussian estimator for μ if*

$$\mathbb{P} \left[|\bar{\mu}(X) - \mu| > L\sigma \sqrt{\frac{\ln(1/\delta)}{n}} \right] \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (3)$$

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clearer

1.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k
Output: Estimate $\hat{\mu}$
 Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;
for $j = 1, \dots, k$ **do**
 Compute the block mean $\bar{\mu}_j = \frac{1}{m} \sum_{i \in B_j} X_i$;
return $\hat{\mu} = \text{med}(\bar{\mu}_1, \dots, \bar{\mu}_k)$;

Theorem 1.4 (MoM concentration). *Let $(X_i)_{i=1}^n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\text{MoM}}$ the MoM estimator with k blocks. Then*

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq e^{-k/8}. \quad (4)$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ for $\delta \in (0, 1)$, this becomes

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \sigma \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta, \quad (5)$$

which is sub-Gaussian in the sense that the dependence on δ is logarithmic.

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}_j] = \mu$ and $\text{Var}(\bar{X}_j) = \sigma^2/m$. By Chebyshev's inequality with threshold $2\sigma/\sqrt{m}$,

$$\mathbb{P} \left[|\bar{X}_j - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq \frac{\sigma^2/m}{4\sigma^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta_j = \mathbf{1}\{|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta_j)_{j=1}^k$ are independent.

If $|\hat{\mu}_{\text{MoM}} - \mu| > 2\sigma/\sqrt{m}$, the median exceeds $\mu + 2\sigma/\sqrt{m}$ or falls below $\mu - 2\sigma/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}$, since the median of k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq \mathbb{P} \left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2} \right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^k \zeta_j$ consists of k independent random variables with $\zeta_j \in [0, 1]$ and $\mathbb{E}[\sum_j \zeta_j] \leq k/4$. For $\sum_j \zeta_j$ to reach $k/2$, it must exceed its mean by

at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P} \left[\sum_{j=1}^k \zeta_j - \mathbb{E} \left[\sum_j \zeta_j \right] \geq \frac{k}{4} \right] \leq \exp \left(\frac{-2(k/4)^2}{k} \right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$ and noting $m \geq n/(2k)$ for $n \geq 2k$. $\square$

References

- Devroye, L., Lerasle, M., Lugosi, G. & Oliveira, R. I. (2016). Sub-gaussian mean estimators.
- Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American statistical association*, 58(301), 13–30.
- Lugosi, G. & Mendelson, S. (2019). Mean estimation and regression under heavy-tailed distributions: A survey. *Foundations of Computational Mathematics*, 19(5), 1145–1190.